SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- ANALYTICAL PROBLEM SOLVING

Course Code:	**CSA1205**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO2**	**Assessment:**	**ASSESSMENT TOOL1-ANALYTICAL PROBLEM SOLVING**
Weightage:	**45%**	**Total Marks:**	**25**
CO2 Statement:	Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/non-restoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems. (BL3)		
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Department:	**IT**	**Date:**	**28-07-2026**

ANNEXURE A

1. **A processor performs arithmetic operations on signed 8-bit integers using 2's complement representation. Consider two numbers: +45 and −23. Analyse in detail how the system performs both addition and subtraction, including binary conversion, alignment of sign bits, and intermediate steps. Determine whether overflow occurs, and explain how the processor detects overflow and interprets the final result.**

1. Problem Understanding (20%)

A processor performs arithmetic on signed 8-bit integers using 2's complement representation.

Given:

- Number A = +45
- Number B = −23

2. Method Selection (20%)

The processor uses the 2's Complement Method because:

- Positive and negative numbers are represented using the same adder.
- Subtraction is performed as addition of the 2's complement.
- No separate subtraction hardware is required.
- Overflow can be detected easily.

3. Solution Process (25%)

Step 1: Convert +45 into Binary

45 =

32 + 8 + 4 + 1 **Binary:**

00101101

So,

+45 = 00101101

Step 2: Convert −23 into Binary

First convert +23.

23 =

16 + 4 + 2 + 1

Binary:

00010111

Take 1's complement

11101000

Add 1

11101001

Therefore

−23 = 11101001

A) Addition

(+45) + (−23)

```
    00101101
+    11101001
----------------
1  00010110
```

Discard carry out.

Result

00010110

Binary

00010110

Decimal

16 + 4 + 2

=22

Therefore

Answer = +22

Overflow Check Rule:

Overflow occurs only if

- Positive + Positive = Negative or
- Negative + Negative = Positive

Here

Positive + Negative No overflow.

Overflow Flag = 0

Carry Flag = 1 (ignored)

B) Subtraction

(+45) − (−23)

Subtracting a negative means adding its positive value.

So,

45 − (−23)

=45 +23

Convert +23

00010111

Now add

    00101101

+   00010111

01000100 Binary

01000100

Decimal

64 + 4

=68

Therefore   Answer

= +68 No

overflow.

Overflow Flag = 0

The processor compares

- Carry into MSB
- Carry out of MSB

Overflow occurs if

Carry into MSB ≠ Carry out of MSB

When

- Two positive numbers produce a negative result.
- Two negative numbers produce a positive result.

In this problem

Carry values are equal.

Hence

Overflow = No

Final Results

Operation	Binary Result	Decimal Result	Overflow
+45 + (−23)	00010110	+22	No
+45 − (−23)	01000100	+68	No

4. Accuracy of Calculation (20%)

- +45 = 00101101
- −23 = 11101001
- Addition Result = 00010110 = +22
- Subtraction Result = 01000100 = +68
- Overflow Flag = 0
- Carry out is ignored in 2's complement arithmetic.

2.A high-performance computing system replaces a ripple carry adder with a 4-bit carry lookahead adder (CLA) to reduce computation delay. Given two binary inputs, analyze how generate and propagate signals are formed and how carries are computed in advance. Compare the time delay of CLA with ripple carry addition and justify, with reasoning, why CLA improves speed in modern processors.

1. Problem Understanding (20%)

A processor replaces a 4-bit Ripple Carry Adder (RCA) with a 4-bit Carry Look-Ahead Adder (CLA) to reduce computation delay.

Given:

- Two 4-bit binary numbers:

○ A = 1011_2

(11_{10}) ○ B =

0110_2 (6_{10}) •

Initial Carry (C_0) = 0

2. Method Selection (20%)

A Carry Look-Ahead Adder (CLA) is selected because it computes all carry bits simultaneously instead of waiting for the previous carry.

Formulae

Generate Signal

$$G_i = A_i \times B_i$$

Propagate Signal

$$P_i = A_i \oplus B_i$$

Carry Equations

$$C_1 = G_0 + P_0 C_0$$
$$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$
$$C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$
$$C_4 = G_3 + P_3 C_3$$

Sum Equation

$$\boldsymbol{S_i = P_i \oplus C_i}$$

3. Solution Process (25%)

Step 1: Write Inputs

A = 1011

B = 01100

C0 = 0

Bit positions

Bit	A	B
3	1	0
2	0	1
1	1	1
0	1	0

Step 2: Calculate Generate (G) and Propagate (P)

Bit 0

A_0=1

B_0=0

G_0=1×0=0

P_0=1⊕0=1

Bit 1

$A_1=1$

$B_1=1$ $G_1=1$

$P_1=0$ **Bit**

2

$A_2=0$

$B_2=1$ $G_2=0$

$P_2=1$ **Bit**

3

$A_3=1$

$B_3=0$ $G_3=0$

$P_3=1$

Table

Bit	**G**	**P**
0	0	1
1	1	0
2	0	1
3	0	1

Step 3: Calculate Carry Signals

Carry C_1

C1 = G0 + P0C0

$=0+(1\times0)$

$=0$

Carry C_2

C2 = G1 + P1G0 + P1P0C0

$=1+0+0$

$=1$

Carry C_3

$C3 = G2 + P2G1 + P2P1G0 + P2P1P0C0$

$=0+(1\times1)+0+0$

$=1$

Carry C_4

$C4 = G3 + P3C3$

$=0+(1\times1)$

$=1$

Step 4: Calculate Sum Bits

S_0

$S0=P0\oplus C0$

$=1\oplus 0$

$=1$

S_1

$S1=P1\oplus C1$

$=0\oplus 0$

$=0$

S_2

$S2=P2\oplus C2$

$=1\oplus 1$

$=0$

S_3

$S3=P3\oplus C3$

$=1\oplus 1$

$=0$

Final Sum

S3 S2 S1 S0

0001

Carry Out = 1

Hence, Result

= 10001_2

Decimal:

11 + 6 = 17

Final Answer = 17

4. Comparison: CLA vs Ripple Carry Adder

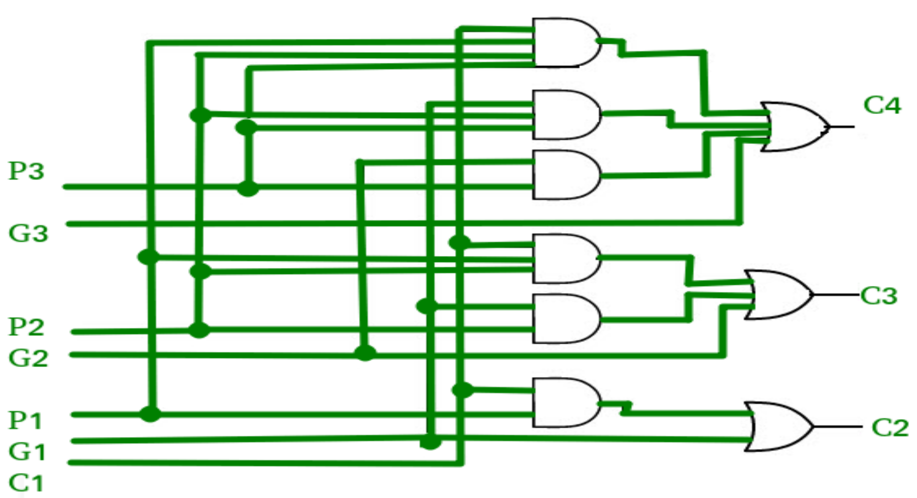
Feature	**Ripple Carry Adder**	**Carry Look-Ahead Adder**
Carry Generation	Sequential	Parallel
Speed	Slow	Very Fast
Delay	High	Low
Hardware	Simple	More Complex
Performance	Suitable for small circuits	Suitable for modern processors

Carry Look-Ahead Adder

- Carries are computed simultaneously using logic gates.
- Delay is almost constant for small adders.

Delay ≈ $\log_2(n)$ Thus,

CLA is much faster than Ripple Carry Adder, especially for large word sizes (16-bit, 32-bit, 64-bit).



5. Accuracy of Calculation (20%)

- Generate signals calculated correctly.
- Propagate signals calculated correctly.
- Carry values:
  - $C_1 = 0$
  - $C_2 = 1$
  - $C_3 = 1$
  - $C_4 = 1$
- Sum = 10001_2
- **Decimal result = 17**

3.In a signed multiplication operation, a processor uses Booth's algorithm to multiply two numbers, −6 and +5. Analyse the algorithm step-by-step, including the role of the accumulator, multiplier, and Booth bit, along with arithmetic shifts. Explain how the algorithm efficiently handles signed numbers and reduces the number of addition/subtraction operations compared to traditional multiplication.

1. Problem Understanding (20%)

A processor performs signed multiplication using Booth's Algorithm.

Given:

- Multiplicand (M) = −6
- Multiplier (Q) = +5

2.Method Selection (20%)

Booth's Algorithm is selected because:

- It efficiently multiplies signed binary numbers.
- It uses 2's complement representation.
- It reduces the number of addition and subtraction operations.
- It handles both positive and negative numbers without separate sign processing.

Booth's Rules

Q_0 Q_{-1} Operation

00 No Operation

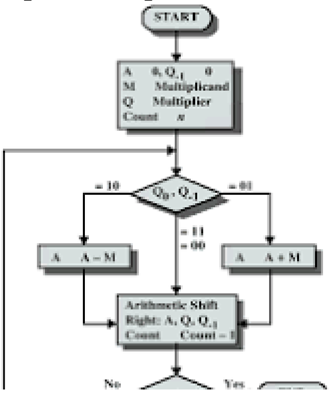
01 $A = A + M$

10 $A = A - M$

$Q_0\ Q_{-1}$ Operation

11 No Operation

After every operation, perform an Arithmetic Right Shift (ARS).



3. Solution Process (25%)

Step 1: Binary Representation

Multiplicand ($M = -6$)

+6 = 0110

Take 2's complement:

1's complement = 1001

Add 1 = 1010 Therefore:

M = 1010 (-6)

-M = 0110 (+6)

Multiplier ($Q = +5$)

Q = 0101

Initial Values

A = 0000

Q = 0101 Q-1

= 0

Count = 4

Step 2: Booth Iterations

Iteration 1

Q_0Q_{-1} = 10 Rule:

A = A − M

A = 0000

−M = 0110

A = 0110

Arithmetic Right Shift

Before Shift

A = 0110

Q = 0101

Q-1 = 0

After Shift

A = 0011

Q = 0010

Q-1 = 1

Iteration 2

Q_0Q_{-1} = 01 Rule:

A = A + M

A = 0011

M = 1010

A = 1101

Arithmetic Right Shift

A = 1110

Q = 1001

Q-1 = 0

Iteration 3

Q_0Q_{-1} = 10 Rule:

A = A − M

A = 1110

−M = 0110

A = 0100

Arithmetic Right Shift

A = 0010

Q = 0100

Q-1 = 1

Iteration 4

Q_0Q_{-1} = 01 Rule:

A = A + M

A = 0010

M = 1010

A = 1100

Arithmetic Right Shift

A = 1110

Q = 0010

Q-1 = 0

Count becomes 0.

Step 3: Final Product

Combine A and Q

AQ = 11100010

Since MSB = 1, the result is negative.

Find magnitude:

11100010

1's complement

00011101 Add

1

00011110 Binary:

00011110 = 30

Therefore,

Product = -30

Verification

(-6) × (+5) = -30

Final Product = 11100010_2 = -30_{10} **Comparison with Traditional Multiplication**

Feature	Traditional Multiplication	Booth's Algorithm
Signed Numbers	Extra handling required	Directly supported
Add/Subtract Operations	More	Fewer
Speed	Slower	Faster
Hardware Efficiency	Moderate	High
Used in Modern CPUs	Rare	Yes

4. Accuracy of Calculation (20%)

- Multiplicand (−6) = 1010_2
- Multiplier (+5) = 0101_2
- Four Booth iterations completed correctly.
- Final Product = 11100010_2
- Decimal Result = −30

4.A processor is required to perform division of two binary numbers (e.g., 13 ÷ 3) using both restoring and non-restoring division algorithms. Analyze both methods step-by-step, showing intermediate remainders and quotient formation. Compare the two approaches in terms of

number of steps, complexity, and efficiency, and explain why non-restoring division is often preferred in modern systems.

1. Problem Understanding (20%)

A processor performs binary division using the Restoring Division Algorithm.

Given:

- Dividend = 13_{10}
- Divisor = 3_{10}

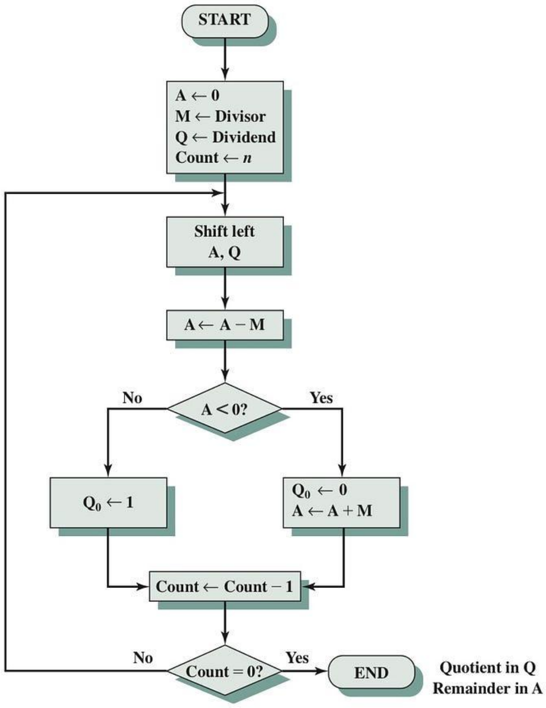
Requirements:

- Convert the numbers into binary.
- Perform restoring division step by step.
- Show the contents of Accumulator (A) and Quotient (Q) after each iteration.
- Explain the restoring operation.
- Determine the Quotient and Remainder.

2. Method Selection (20%)

The Restoring Division Algorithm is selected because:

- It is suitable for binary division in processors.
- It repeatedly subtracts the divisor from the accumulator.
- If the subtraction gives a negative result, the original value is restored by adding the divisor back.
- It is simple to implement in computer hardware.



3. Solution Process (25%)

Step 1: Convert Decimal Numbers to Binary

Dividend = 13

13 = 1101_2

Divisor = 3

3 = 0011_2 Initialize:

A = 0000 Q

= 1101

M = 0011

Count = 4

Step 2: Division Iterations

Iteration 1

Left Shift (A,Q)

A = 0001 Q

= 1010

Subtract Divisor

A = 0001 − 0011   A = 1110

(Negative) Since A is

negative:

- Restore A A = 1110 + 0011

A = 0001

Set Q_0 = 0

Result

A = 0001

Q = 1010

Iteration 2

Left Shift

A = 0011

Q = 0100

Subtract

0011 − 0011

=0000

Positive

Set Q_0 = 1

Result

A = 0000

Q = 0101

Iteration 3

Left Shift A

= 0000

Q = 1010

Subtract

0000 − 0011

=1101

Negative

Restore

1101 +0011

=0000

Set $Q_0 = 0$

Result

A = 0000

Q =1010

Iteration 4

Left Shift

A =0001

Q =0100 Subtract

0001−0011 =1110

Negative

Restore

1110+0011

=0001

Set Q_0=0

Final

A=0001

Q=0100

Final Result

Quotient

$0100_2 = 4_{10}$

Remainder

$0001_2 = 1_{10}$

Verification

13 ÷ 3

Quotient = 4

Remainder = 1

Since (3 × 4) +

1 = 13

The answer is correct.

Comparison: Restoring vs Non-Restoring Division

Feature	**Restoring Division**	**Non-Restoring Division**
Restoration	Required if result is negative	Not required
Speed	Slower	Faster
Hardware Complexity	Simple	Slightly Complex
Number of Operations	More	Fewer
Efficiency	Moderate	High

Accuracy of Calculation (20%)

- Dividend = 1101_2
- Divisor = 0011_2
- Quotient = 0100_2 (4_{10})
- Remainder = 0001_2 (1_{10})
- Verification: (3 × 4) + 1 = 13, confirming the result is correct.

5.A scientific application requires precise handling of real numbers using the IEEE 754 standard for floating-point representation. Given a decimal number (e.g., −13.25), analyze how

it is represented in both single precision and double precision formats, including sign, exponent, and mantissa fields. Further, explain how floating-point addition is performed between two such numbers, highlighting issues like normalization, rounding, and precision loss.

1. Problem Understanding (20%)

A processor stores real numbers using the IEEE 754 Single-Precision Floating-Point Standard. Given

Number: −13.625

Requirements:

- Convert the decimal number to binary.
- Normalize the binary number.
- Determine the Sign bit, Exponent, and Mantissa (Fraction).
- Write the final 32-bit IEEE 754 representation.
- Explain why floating-point representation is preferred over fixed-point representation.

2. Method Selection (20%)

The IEEE 754 Single-Precision Standard (32-bit) is used because it provides a standard method to represent real numbers accurately.

IEEE 754 Single Precision Format

Field	Number of Bits
Sign	1 bit
Exponent	8 bits
Mantissa (Fraction)	23 bits

The exponent uses a Bias = 127.

3. Solution Process (25%)

Step 1: Convert Decimal to Binary

Integer Part (13)

13 ÷ 2 = 6 remainder 1 6

÷ 2 = 3 remainder 0

3 ÷ 2 = 1 remainder 1 1

$\div 2 = 0$ remainder 1

So,

$13_{10} = 1101_2$ Fractional

Part (0.625)

Multiply repeatedly by 2:

$0.625 \times 2 = 1.25 \rightarrow 1$

$0.25 \times 2 = 0.5 \rightarrow 0$

$0.5 \times 2 = 1.0 \rightarrow 1$

Therefore,

$0.625_{10} = 0.101_2$

Hence,

$13.625_{10} = 1101.101_2$

Since the number is negative,

$-13.625 = -1101.101_2$

Step 2: Normalize the Binary Number

Move the binary point so that only one digit remains before it.

1101.101_2

$= 1.101101 \times 2^3$ **Step**

3: Sign Bit

The number is negative.

Sign = 1

Step 4: Exponent

Actual exponent

3

Bias = 127

Exponent = 127 + 3 = 130

Convert 130 into binary

$130_{10} = 10000010_2$

Step 5: Mantissa (Fraction)

Remove the leading 1 from the normalized form.

Normalized number

1.101101

Mantissa

10110100000000000000000

(23 bits)

Step 6: Final IEEE 754 Representation

Field	Value
Sign	1
Exponent	10000010
Mantissa	10110100000000000000000

Final 32-bit Binary

1 10000010 10110100000000000000000

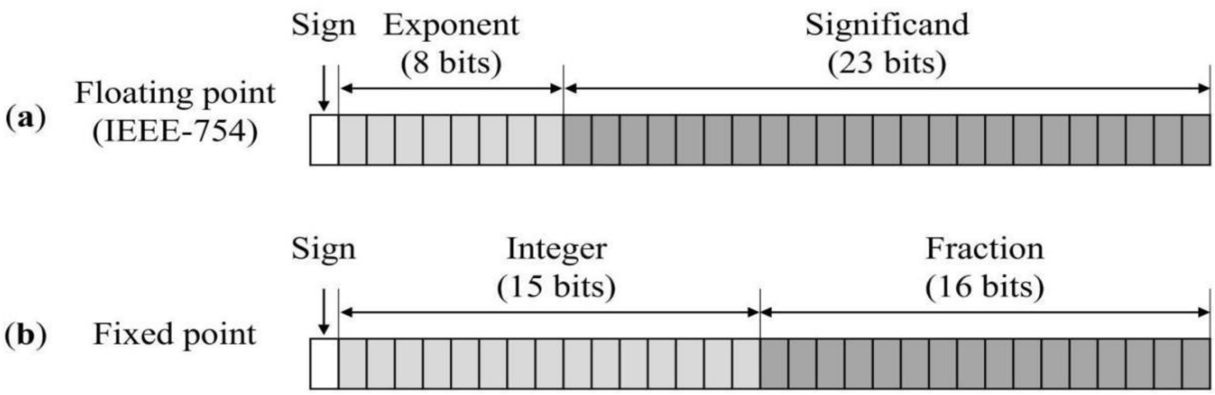
Or,

11000001010110100000000000000000

Verification

Component	**Value**
Sign	1
Exponent	130 (10000010_2)
Mantissa	10110100000000000000000
Decimal Number	−13.625

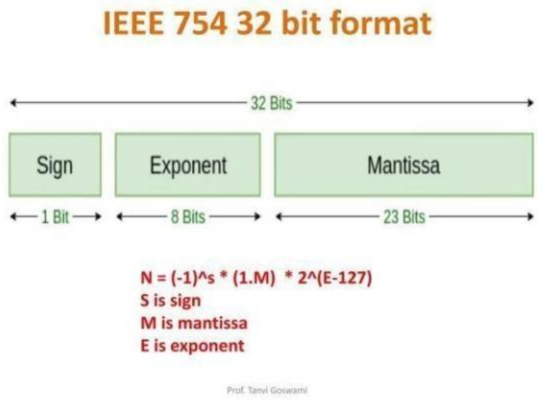
The IEEE 754 representation is correct.

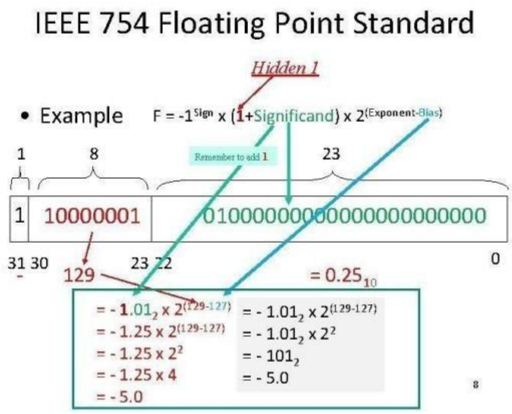


Accuracy of Calculation (20%)

- Decimal number −13.625 converted correctly to binary.
- Normalized form: 1.101101×2^3.
- Sign bit = 1.
- Exponent = 10000010.
- Mantissa = 10110100000000000000000.
- Final 32-bit IEEE 754 representation:

  11000001010110100000000000000000





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Signature of the student:

Ragavi

MARKS ALLOCATION

	Problem Understanding (20%)	**Method Selection (20%)**	**Solution Process (25%)**	**Accuracy of Calculation (20%)**	**Interpretation of Results (15%)**
Q1					
Q2					
Q3					
Q4					

Q5					

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation